Machine Learning: k-Means and PCA for Regime Detection

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1. Introduction

2. Data Processing and Feature Engineering

This project focuses on regime detection on multi-dimensional financial time-series. We track 9 major stock indices across 3 different regions (America, Europe, Asia) to develop our analysis.

Table 1: Index universe by region.

Region	Index	Ticker
America	S&P 500	`^GSPC`
America	Dow Jones	`^DJI`
America	Nasdaq	`^IXIC`
Europe	FTSE 100	`^FTSE`
Europe	CAC 40	`^FCHI`
Europe	DAX	`^GDAXI`
Asia	KOSPI	`^KS11`
Asia	Nikkei 225	`^N225`
Asia	Hang Seng	`^HSI`

2.1. Raw Data Structure

Each index is downloaded from Yahoo Finance with daily OHLCV columns: `Date`, `Open`, `High`, `Low`, `Close`, and `Volume`. After merging, the full DataFrame uses a MultiIndex on columns of the form `(index_name, OHLCV_field)`, so that, for example, S&P 500 closing prices are accessed via `df["SP500"]["Close"]`.

2.2. Feature Engineering

From the closing prices two families of features are computed.

Log-returns. Over multiple horizons (default: 1, 5, and 21 trading days):

$$r_{t,p} = \ln\left(\frac{P_t}{P_{t-p}}\right). \tag{1}$$

Rolling annualised volatility. Default windows of 21 and 63 days:

$$\sigma_{t,w} = \mathrm{std}(r_{t-w:t}) \times \sqrt{252}\,. \tag{2}$$

This produces a feature matrix of shape $(T,\ 9 \times (3 \text{ returns} + 2 \text{ volatilities})) = (T, 45)$ columns, where each row corresponds to one trading day and each column to a pair `(index_name, feature_name)`.

In order to capture the dynamic nature of regime, features are computed in a rolling window fashion, so that the feature vector at time t is based on data from the previous w days. For example, the rolling correlation between two indices X and Y at time t with window w is computed as:

$$\rho_{X,Y,t} = \frac{\sum_{i=0}^{w-1}(r_{X,t-i} - \bar{r}_X)(r_{Y,t-i} - \bar{r}_Y)}{\sqrt{\sum_{i=0}^{w-1}(r_{X,t-i} - \bar{r}_X)^2 \ \sum_{i=0}^{w-1}(r_{Y,t-i} - \bar{r}_Y)^2}} \tag{3}$$

where $\bar{r}_X$ and $\bar{r}_Y$ are the mean returns of indices X and Y over the window w.

We additionally compute momentum Z-score features to measure mean-reversion by normalising the distance between the current price and the moving average by the rolling volatility:

$$Z_{t,w} = \frac{P_t - \mathrm{MA}_{t,w}}{\sigma_{t,w}} \tag{4}$$

2.3. Pipeline Summary

Yahoo Finance API

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Raw OHLCV (per index)

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Merged DataFrame (MultiIndex columns)

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Close prices only (simple DataFrame)

↓

Log-returns (1d, 5d, 21d) Rolling volatility (21d, 63d)

↓

Feature matrix → Clustering / PCA

Figure 1: Data pipeline from raw download to feature matrix.

3. Mathematical Foundations

In this project we apply two main machine learning techniques for regime detection: k-means clustering and Principal Component Analysis (PCA).

3.1. K-Means Clustering

K-means clustering is an unsupervised learning algorithm that partitions the data into K clusters by minimising the within-cluster sum of squares. The algorithm iteratively assigns each data point to the nearest cluster centroid and then updates the centroids based on the assigned points.

The Within-Cluster Sum of Squares (WCSS) is defined as:

$$\text{WCSS} = \sum_{k=1}^{K} \sum_{x_i \in C_k} \|x_i - \mu_k\|^2 \tag{5}$$

where C_k is the set of points in cluster k and μ_k is the centroid of cluster k.

However, when dealing with higher-dimensional data, the standard k-means algorithm can be limited due to the curse of dimensionality as well as the different topologies of the clusters. To address this, one can reduce dimensionality using PCA before applying k-means.

3.2. Principal Component Analysis (PCA)

4. Proposed Algorithm

We propose the following algorithm for regime detection:

Algorithm 1 K-Means Regime Detection Pipeline

Require: Closing price matrix $\mathbf{P} \in \mathbb{R}^{T \times N}$ for N indices over T trading days
Require: PCA variance threshold $\alpha \in (0, 1]$, candidate range $[K_{\min}, K_{\max}]$
Ensure: Regime labels $\ell_t \in \{0, \ldots, K^* - 1\}$ for each day t, ordered by mean return

— **Stage 1: Feature Engineering** —
for each index $n = 1, \ldots, N$ **do**
  for each horizon $p \in \{1, 5, 21\}$ **do**
    $r_{t,p}^{(n)} \leftarrow \ln(P_t^{(n)} / P_{t-p}^{(n)})$ ▷ Log-returns
  end for
  for each window $w \in \{21, 63\}$ **do**
    $\sigma_{t,w}^{(n)} \leftarrow \mathrm{std}(r_{t\text{-}w:t}^{(n)}) \times \sqrt{252}$ ▷ Rolling volatility
  end for
  $Z_t^{(n)} \leftarrow (P_t^{(n)} - \mathrm{MA}_{t,63}^{(n)}) / \sigma_{t,63}^{(n)}$ ▷ Momentum z-score
  $\mathrm{RSI}_t^{(n)} \leftarrow \mathrm{RSI}_{14}(P^{(n)})$ ▷ Relative Strength Index
end for
for each pair (i, j) with $i < j$ **do**
  $\rho_t^{(i,j)} \leftarrow \mathrm{RollingCorr}_{63}(r^{(i)}, r^{(j)})$ ▷ Pairwise correlation
end for
Concatenate all features into $\mathbf{F} \in \mathbb{R}^{T' \times D}$ and drop warm-up rows

— **Stage 2: Standardisation** —
for each feature $d = 1, \ldots, D$ **do**
  $\tilde{F}_{t,d} \leftarrow (F_{t,d} - \hat{\mu}_d) / \hat{\sigma}_d$ ▷ Z-score normalisation
end for

— **Stage 3: Dimensionality Reduction** —
$\mathbf{U}, \mathbf{\Sigma}, \mathbf{V}^\top \leftarrow \mathrm{SVD}(\tilde{\mathbf{F}})$
Select $d^* = \min\{d : \sum_{i=1}^{d} \sigma_i^2 / \sum_{i=1}^{D} \sigma_i^2 \geq \alpha\}$
$\mathbf{X} \leftarrow \tilde{\mathbf{F}}\, \mathbf{V}_{:,1:d^*} \in \mathbb{R}^{T' \times d^*}$ ▷ Project onto top d^* PCs

— **Stage 4: Optimal K Selection** —
for $K = K_{\min}, \ldots, K_{\max}$ **do**
  Run K-Means on $\mathbf{X}$ with K clusters
  Record inertia $\mathcal{I}(K)$ and silhouette score $\mathcal{S}(K)$
end for
$K^* \leftarrow \arg\max_K \ \mathcal{S}(K)$ ▷ Best silhouette

— **Stage 5: Final Clustering** —
$\hat{\ell}_1, \ldots, \hat{\ell}_{T'} \leftarrow \mathrm{KMeans}(\mathbf{X},\ K^*)$

— **Stage 6: Regime Ordering** —
for each cluster $k = 0, \ldots, K^* - 1$ **do**
  $\bar{r}_k \leftarrow \mathrm{mean}(\{r_t : \hat{\ell}_t = k\})$ ▷ Mean basket return
end for
$\pi \leftarrow \mathrm{argsort}(\bar{r}_0, \ldots, \bar{r}_{K^*-1})$ ▷ Rank worst → best
$\ell_t \leftarrow \pi^{-1}(\hat{\ell}_t) \quad \forall\, t$ ▷ Relabel: 0 = bearish, $K^* - 1$ = bullish

return $\boldsymbol{\ell} = (\ell_1, \ldots, \ell_{T'})$

5. Results

6. Conclusion